

Boric Alchemy Documentation:

Foreword: Hey, hey, it's ya' boy, MrBadWithNames!

I have recently been asked for some form of documentation, and since the handbook is a nightmare to edit and update (because it's one long line of text, interspersed with vintage story's HTML format.)

Put yourself in my shoes, and try to edit:

"boricalchemy:metals_text": "Aren't metals great? Wouldn't you like more of them?
Sure, maybe the crude mitts of a soot-stained blacksmith struggle to collect every last bit of metal from their ore – but not us alchemist.

By definition, ore is impure, so what works on one might not necessarily work on another.

Typically, ores are dissolved in an acid, forming a salt that is later reduced back into metal bits.
There are three main tiers of metal salts, each requiring charcoal to turn back into metal.

Sulfates are the roughest tier, giving only four nuggets per crushed ore.
Nitrates require only one litre of acid instead of two and give five nuggets when reduced.
Chlorides also only require one litre of acid and produce six nuggets when reduced; however, they also require potash on top of the regular charcoal.

Sulfuric acid can transform the following:
Crushed copper into Copper Sulfate,
Crushed iron into Iron Sulfate,
Crushed lead into Lead Sulfate,
Crushed nickel into Nickel Sulfate,
Crushed zinc into Zinc Sulfate,
Crushed chromium into Chromium Sulfate,

Nitric acid can transform the following:
Crushed copper into Copper Nitrate,
Crushed iron into Iron Nitrate,
Crushed lead into Lead Nitrate,
Crushed nickel into Nickel Nitrate,
Crushed zinc into Zinc Nitrate,

Hydrochloric acid can transform the following:
Crushed copper into Copper Chloride,
Crushed iron into Iron Chloride,
Crushed bismuth into Bismuth Chloride,
Crushed nickel into Nickel Chloride,
Crushed zinc into Zinc Chloride,
Crushed chromium into Chromium Chloride,

Tin Chloride,

Auqua regia is a mix of nitric and hydrochloric acid capable of dissolving gold with ease and silver with a little effort!
Crushed gold is turned into Gold Chloride,
Crushed silver is turned into Silver Chloride,

For more advanced extraction of gold and silver, Cyanide can be used to create:
Potassium Cyanide Silver,

For maximal yield, Mercury can be used to form:
Silver-Mercury Amalgam,
or Gold-Mercury Amalgam,

Tin is fairly immune to most acids; however, with a bit of potash and some crushed cassiterite, you can make:
Potassium Stannate which can be processed back into tin nuggets!
"

Horrible, terrible – ask the wind for recipes instead, this shit would make me head-butt a unicorn until into a lobotomy.

So – I wrote, and am writing, this.

Expect typos – and unless the typo is a fucking slur or complexly unrecognizable – I'm probably not going to be fucked to fix it. 😊

Okay? Okay!

Also – this is not a copy of the handbook – I won't re-explain every single recipe, especially for recipes with many variants.

Also, in-game, I like to keep a *narrator tone* – expect more swearing.

Hydrocarbons.

While I won't explain every distillation in detail, if petroleum is important enough to murder each other in the real world, I suppose it's only fair I explain it a little in the context of Boric Alchemy.

Oily Rocks are functionally an ore. They are deposits of crude oil squeezed into porous rocks, like limestone, shale, or sandstone.

These rocks must be crushed and fired in a **retort** to get petroleum.

Petroleum must be run through a fractional distiller to extract the fractions within.

As of writing this document, these fractions are:

Gasoline (good solvent, fuel for flamethrowers and grenades),

Naphtha (feedstock for plastic alcohols and carboxylic acids),

Diesel (very energy-dense fuel),

Carbolic Oil (can be made into phenol, cresols, or nepheline. Useful for synthetic dye),
Mineral Oil (can replace fat in many vanilla recipes – but not the one where it’s supposed to burn),
Paraffin Wax (can replace bees wax in many vanilla and modded recipes)

Making Resin.

Tree resin is used for a bunch of things, from mechanical machinery, to wallpaper adhesive.
A solvent, like **gasoline, acetone, or benzene** can be used to extract resin from crushed pine or acacia wood.

You can also make 100% chemical resin out of formaldehyde (as the solvent) and lignin (extracted from sawdust), or phenol and phosphoric acid.

NOTE: lignin and charcoal can also be mixed directly in a pot to make pitch glue!

Glass Jug.

Use: a way to trick vintage story into letting you craft a liquid output in the grid >:).

Variants: empty, benzyl chloride, glycerine, hcl, h2o2, hypochlorous acid, sulfuric acid. (there are also other kinds that work like light sources)

Note: if you ever can’t find a recipe for a liquid, look for it in a jug.

Autoclaves.

Use: similar to how a retort/fractional distiller works – they are filled in the grid, cooked in a fire pit – but unlike the retorts, **the autoclave is emptied in the grid** – not by clicking on it.

Variants: doped quartz, aquamarine, red/blue/fire clay, emeralds, clear quartz, peridot, rose quartz, polyethylene fluff,

Recipes: the autoclave itself must be crafted from steel, tools, and soldering stuff.

Note: take a shot every time you read “soldering stuff”

Finished Wood.

Use: Nicer, more saturated wood.

Variants: all vanilla wood types (not modded wood types – these aren’t shaders, that’s a new block, those don’t appear by themselves)

Recipes: wood block in a barrel of wood finish.

Note: works for stairs and slabs – and can be sawed back into regular planks

Anthills.

Use: if broken, they drop ants that can be eaten or crushed and distilled into formic acid.
Formic acid is needed to make speed.

Crafted: from any fertility/coverage dirt, a bit of food (they love sugar) and some ants.

Note: these are farmable if you craft a forming anthill.

Bonfire.

Use: when it, it melts ice and snow, and cracks rocks and ores, making them require 1 tool tier less than usual.

Crafted: firewood and saltpetre (and oxidiser)

Note: highly configurable :D

Iron Cauldron.

Use: can hold more liquids and solids

Crafted: smithed from iron (or cast from cast iron with Wulk's smithing additions)

Glazed Storage Vessel.

Use: colour storage vessels

Crafted: first combine any raw storage vessel with water (bucket) and a metal salt (like copper chloride) These wet storage vessels must be fired.

Variants: Green, Red, Yellow, Blue, Black, Orange, Teal, White, and Shulker Chest Purple :p

Other Pottery.

- **Beaker:** for fractional distillers, can hold 3 litres of any liquid like a bigger bowl.
- **Quartz-Lined Chemical Vessel:** clayformed, lined with crushed quartz, and only THEN fired – it can store more chemicals than a full barrel (assuming a litres per block calculation)

Hydrothermal Vents.

Use: found on the sea floor, the vents can be broken and left to re-grow for a renewable source of metal salts and sulfur.

Note: if broken in their forming stage, they will not re-grow.

Cobalt Rich Crust.

Use: found on the sea floor, the crust spawns in a saltpetre-like layer.

Note: this will not "re-grow" in your lifetime (like at all, but irl it'd take AAAGES).

Tungsten Cube.

Use: desk ornament.

Note: pretty cool tho.

Recipe: bowls, and tungsten powder, thermite, and clay. You seal it – cook it – crack it with a hammer – and get your cube out!

Metal Salts.

- **Aqueous Metal Sulfates** = Crushed Metal + Sulfuric Acid
Dry Metal Sulfate = aqueous sulfate dried in a barrel.
Can be reduced into metal bits or used for specific recipes.
- **Aqueous Metal Nitrates** = Crushed Metal + Nitric Acid
Dry Metal Nitrate = aqueous nitrate dried in a barrel.
Can be reduced into metal bits or used for specific recipes.
- **Aqueous Metal Chloride** = Crushed Metal + HCL Acid
Dry Metal Chloride = aqueous chloride dried in a barrel.
Can be reduced into metal bits or used for specific recipes.

Chlorides have better yields than nitrates.

Nitrates have higher yields than sulfates.

Sulfates have higher yields than a hammer.

Many metal salts can mordant cloth, work as a pigment, or be involved in a more efficient displacement reaction at the cost of a more reactive metal!

Ceramic Retorts.

Use: retorts can be filled and fired in a kiln (pit or beehive) to destructively distil chemicals.

Variants: activated charcoal, mercury, coke, petroleum, sodium carbonate, acetone, white/red phosphorus, formaldehyde.

Recipes: first clayformed and fired empty like any ceramic, the retort is re-usable and must be right-clicked to harvest the content within. To load it up – just use the grid.

Note: the handbook doesn't connect the retort to its yields – read item descriptions if in doubt!

Fractional Distiller.

Use: Extract several liquids out of the distillers contents in a firepit.

Variants: ale, black coal, carbolic oil, wood, petroleum, 2 naphthas,.

Recipes: copper, soldering tools, Beakers have to be added to it in the grid one at a time.

Note: the handbook doesn't connect the distiller to its yields – read item descriptions if in doubt!

Some Decor.

- **Zinc Sulfide Starts:** emissive wall decals, comes in skull, gear, heart, and star shapes!
Crafted: wax + zinc sulfide

- **Alchemy Papers:** many decals that you can switch through in the grid for free.
Crafted: paper and charcoal
- **Coloured Lights:** Dynamic and placeable coloured lighting.
Crafted: Oil lamp + colourant salt (like lithium for red, or salt for yellow)
- **Coloured Sparklers:** Flashier, smokier, laggier light sources.
Crafted: Coloured Oil lamp + blasting powder
Note: 2 special sparklers exists, a trans one and a rainbow one ^^

Banana Bread.

Use: Eatable, slightly more filling than regular bread and lasts longer.

Variants: dough, part-baked, perfect, and charred.

Recipes: crafted from a bucket of banana-flavoured preservatives and flour base-game flour.

Note: cooked in the clay oven like regular bread.

Mucky Straw.

Use: After 10 days placed in the world, becomes crusty and when broken it drops saltpetre!

Recipes: hay block, potash (craftable from ash), and rot.

Custom Coloured Glass.

Use: pretty :3

Variants: orange (uranium), blue-green (iron salt), dark blue (cobalt oxide),

Recipes: quartz + colourant in a bloomery.

Recipes for Vanilla Coloured Glass.

Use: pretty too :3

Variants: every vanilla glass colour can be made same as the custom colours.

Recipes: quartz + colourant in a bloomery.

Recipes for Rainbow Glass.

Use: oh-so very pretty also :3

Variants: cool (mostly blue), cidery (mostly orange-yellow), nebulous (mostly pink), and lush (mostly green).

Recipes: mortar and pestle + 2 of the main glass (blue, purple, orange, or green) and 1 block of any other colours - subsequent glass-batch placed in a bloomery.

NOTE: for example, a cool rainbow glass batch is crafted from **2 azure glass blocks, a mortar and pestle, and 1 of any OTHER glass.**

Wind-Up Light.

Use: a temporary light source that needs to be wound-up (with a right click in-world) to power them (for a few in-game hours at a time).

Variants: Co2 (blue), Hydrogen (purple), Krypton-Mercury-Argon (white and lasts long), mercury (green), Sodium (yellow, very bright), Xenon (white), Neon (red).

Recipes: wind up dynamo, wires, transformer, gas container, boards, soldering stuff.

Double-Basins.

Use: A multi-step, lengthy ice-making process using water and saltpetre.

Variants: empty, still freezing, ready to harvest, already melted, and dry.

Recipes: > The empty double basin is crafted from a clay pot, boards, nails and strips, and a hammer.

> The empty double basin can be filled with water and saltpetre.

> The freezing double basin will turn into a frozen double basin with ice.

> If you wait too long, the ice will melt and you'll have to salvage your saltpetre.

> If you harvest the ice on time you'll still get your saltpetre back.

Note: the ice is required in some recipes (like making hydrogen peroxide), the ice you get is lake ice.

Advanced Batteries.

Use: Usable in recipes for electrochemistry (any recipe using an electricity source and electrodes).

Variants: > Lead & Lead Dioxide (Charged and Dead).

> Zinc & Silver Chloride (Charged and Dead).

> Zinc & Copper (Charged and Dead).

Recipes: crafted from a lead-battery casing (made from lead plates, tools, and soldering items) + the relevant salts, metals, and or electrolyte.

Note: after being depleted, the battery can be disassembled to get the lead casing and the occasional salt or metal back. (if the chemistry and game engine limitations allow it)

Primitive Batteries.

Use: Usable in recipes for electrochemistry (weaker and less durable than the advanced batteries).

Variants: > Zinc & Sulfur.

> Lead & Sulfur.

Recipes: crafted from the necessary salt, metal, and empty battery cells (crafted from clay bowls).

Note: When depleted, the primitive batteries will transform into empty battery cells.

Alchemist Hat.

Use: Wearable, Warm, and Stylish!

Variants: regular and rubber lined for extra water resistance!

Recipes: crafted from rope, red cloth, and twine.

Note: N/A.

Diving Equipment.

Use: A two-part diving set!

Parts: diving helmet, and diving tank.

Variants: The tank comes in full and empty variants and must be re-filled with a pump if durability runs out.

Recipes: brass plates, glass, rubber seals, nails and strips, and tools.

Note: N/A.

Gas Mask.

Use: Wearable, prevents nearby fumigators (AOE gas damage blocks) from hurting the wearer!

Variants: N/A

Recipes: crafted from wax, cloth, rope, bowls, and activated charcoal (made in a retort with charcoal and water).

Note: Currently it does not lose durability but it may in the future.

Bronze Necklaces.

Use: Wearable, a way to show off gems, especially the ones only an alchemist can obtain!

Variants: Peridot, Emerald, Diamon, Amethyst, Aquamarine (grown in the autoclave), and Topaz.

Recipes: crafted from a gem, tools, a tin bronze chain, and soldering stuff.

Note: Currently it does not lose durability but it may in the future.

Gas Compressor:

Use: Used as part of the atmosphere-liquifying apparatus – used to obtain liquid air.

Recipes: iron, tools, rubber tools, soldering stuff.

Gas Cooler:

Use: Used as part of the atmosphere-liquifying apparatus – used to obtain liquid air.

Recipes: copper, tools, rubber tools, soldering stuff.

Gas Decompressor:

Use: Used as part of the atmosphere-liquifying apparatus – used to obtain liquid air.

Recipes: iron, tools, rubber tools, soldering stuff.

Electrodes:

Use: Used for electrochemical recipes.

Recipes: the metal, salt, and tools.

Variants: copper, iron, lead, lead dioxide, graphite, amorphous carbon,

Filled Crucibles:

Use: electrolysis in molten metal – for example to get sodium metal out of molten salt. First you craft them in your inventory, then you cook them on a fire and use them in the recipe before they cool back down.

Recipes: the crucible and the thing to melt. Can make sodium, aluminium, and beryllium.

Variants: salt, sodium metal, and alumina + cryolite

Electron Accelerator:

Use: Ionize stuff – turn doped quartz (grown in an autoclave) into amethyst, or smoky quartz, or turn a container of oxygen into ozone.

Recipes: steel, lead, copper, tools, induction coils, electron tubes.

Note: for crafting, it must be crafted with a running generator to power it.

Induction Coil:

Use: Electrical equipment and transformer.

Recipes: iron, rubber, wires, and tools.

Polyethylene:

Use: synthetic fibre – used for bags, bows, clothes and so on!

Recipes: > in an autoclave, dry silica gel (doped with Chromium Vi oxide) + ethylene gas makes polyethylene fluff.

> polyethylene fluff cooked in a pot makes polyethylene fibres.

> the fibres make a twine, the twine make a cloth. Can replace flax in most vanilla recipes.

Alchemical Meat Salts:

Use: curing meat to make it last longer, look redder, and taste... longer!

Recipes: salt, saltpetre, and sodium phosphate.

Note: to actually cure some meat, put the powder in a barrel with raw red meat, fish, or poultry.

Candy:

Use: calories and the occasional bit of healing or damage!

Recipes: honey / sucrose + flavouring / dry / milk powder (depending on the kind)

Variants: honeydrops, herbal, banana, toffee, cherry, cyanide (hurts), gasoline (hurts and can burn)

Amphetamine:

Use: speed buff!

Recipes: hcl + formetorex (lotta stuff bro)

Note: have fun ^^

Sulfur Bacteria Incubator:

Use: a way to farm sulfur at the cost of sulfur rich foods like eggs and cabbages.

Recipes: clay pot, copper, oil lantern (for heating), and alcohol to sterilize.

Note: you have to find a piece of the slime to cultivate more, and if you forget about the incubator, it can spoil you can lose your harvest.

Combustion Engine:

Use: a way to turn a fuel into electricity, the engine always has 75 uses, but depending on the fuel it could take 2 or 10 litres of it.

Recipes: copper, magnetite nugget, wires, gears (iron or rusty), rubber, and the assembled engine body (smithed then assembled (partly on the ground in-world, and partly in the grid. If in doubt check the item description).

Special Bandages:

Use: a way to heal better than with just dry cloth.

Recipes: a clean bandage + the item or more commonly liquid needed.

Variants: Epsom salts solution, disinfecting (from phenol or cresols)

Mortar and Pestles:

Use: a low-durability quern / pulveriser alternative for early game with lower yields.

Recipes: hard rock, and tools..

Variants: All the hard rocks like granite and sandstone.

Synthetic Dyes:

Use: add the dry stuff to water or a solvent and get your dye – a concentrated dye has to be watered down first.

Recipes: oxides, hydroxides, and organic chemicals – a lot of things can be dyes if you try hard enough.

- **RED:** alizarin, litharge, potassium dichromate
- **BLUE:** copper hydroxide, concentrated dye
- **PINK:** concentrated dye (phenol derived)
- **WHITE:** lead hydroxide
- **GREEN:** concentrated dye
- **BLACK:** plattnerite
- **ORANGE:** vanadium pentoxide, concentrated dye
- **PURPLE:** mauvine, concentrated dye
- **YELLOW:** dandelions, massicot, picric acid, bismuth III oxide

NOTE: concentrated dyes are usually made with a little water or other solvent (this makes the recipe cheaper, you probably don't want to use 10L of aqua vitae for 10L of green dye you could make out of grass and water :/)

Alchemist Supply Sack:

Use: can hold a bunch of powders and salts.

Recipes: a regular sack, red dye (in bucket), and lead plates.

Steam Powered Engine:

Use: a temporary electricity source has to be fueled by water and coal.

Recipes: from copper and rubber seal,

Soap:

Use: can clean a pot together with sand. Can wash dyed cloth back to mordant cloth.

Recipes: rendered fat and potash – glycerine can be collected if an empty glass jug is included in the recipe.

Ammonia:

Use: main use is bleaching dyed cloth back to plain cloth.

Recipes: rendered fat and potash – glycerine can be collected if an empty glass jug is included in the recipe.

Thermite:

Use: some of the most extreme sources of chemical heat, used to smelt tungsten. DON'T destroy your crucible trying to melt iron!

Recipes: aluminium powder, some metal oxides, and in one case manganese powder.

Rubber:

Use: many recipes,

Recipes: cooking latex with sulfur and charcoal. Latex is extracted from dandelion sap with an acid.

Fertilizers:

Use: better plants.

Variants: ammonium chloride, monocalcium sulfate, land, sea, and bone ash.

NOTE: these are better than vanilla fertilizers usually. Land ash can be turned into potash, marine ash is made from burning "fresh marine weeds", and item dropped by seaweed and sea grass

Slow Burning Briquette:

Use: after being made it is wet and must be dried. Then it can burn for a long time as moderate heat – perfect for warming your home in the winter.

Recipes: ingot mold (not consumed), black coal / charcoal powder, raw clay.

Coloured Paper:

Use: used to craft wallpapers.

Recipes: coloured papers and dilute resin.

NOTE: all vanilla wallpapers can be crafted

Gas Containers:

Use: used in recipes, those that are flammable can be used as fuel (and an empty gas tank will be left over).

Full containers can be emptied into “air” ones.

Air containers can be emptied into “vacuum” ones with mercury or a hand pump.

Recipes: glass, copper, and rubber seals

Weapons:

- **Flamethrower:** hurts and ignites your foes, different fuels can power it for longer or shorter, alcohols are the worst fuels and the specifically formulated Flammol is the best.
Crafted: from iron, empty gas containers, rubber seals, a candle, and tools (this makes the dry flamethrower that must be fueled)
Note: the flamethrowers vary in durability, *not* raw damage.
- **Alchemical Hand Mortar:** a gun, it only has 4 shots before it must be reloaded and the recoil hurts you with every shot. It can crack stones and set off bombs.
Crafted: is secret. But, if you find the ruined hand mortar in a chemical vessel (from the underground lab ruins) or just look the ruined hand mortar up in the handbook if you're playing on homosapiens or are lazy.
Its item description tells you exactly which items you must blindly put together in the grid (shapeless recipe)
Fuel mix A: copper sulfate, potassium permanganate, h₂o₂, sodium silicate.
Alt Fuel Mix: calcium chloride, potassium permanganate, h₂o₂, sodium carbonate.
Note: this weapon is yanked from my webnovel (seeking plagues), 2 of the chemicals mix to form a solid projectile, the other two react and produce a shit load of gas to propel the projectile.
- **Capacitor Spear:** a melee weapon that can be recharged with a generator and an induction coil. It does AOE damage and knocks your targets backwards away from you. It only has 4 hits before needing to be reload.

Crafted: from sticks, capacitors, rubber, and wires (makes the dead variants that must be charged)

- **Impact Grenade:** when thrown, explodes as soon as it hits a target or solid object.
Recipe: jug, rope, picric acid, and acetone peroxide
- **Fuse Grenade:** when thrown, explodes after 1, 2, or 3 seconds, depending if 1, 2, or 3 pieces of fuse are used in the recipe.
Recipe: jug, rope, blasting powder, and fuse
- **Fire Grenade:** when thrown, sets entities around where it lands on fire (does NOT ignite blocks – I'd love to make it, but I can't. I don't know how.).
Recipe: jug, rope, pretty much any liquid fuel that can burn.
- **Fumigator:** deals AOE damage for a while when places, after a while it becomes a dirty pot.
Crafted: pot, sulfur, and a torch. Or a pot and chlorine gas.
- **Mines:** regular and upgraded (upgraded mines break blocks, the regular one doesn't) They explode when touched by any entity.
Crafted: bowls and explosives (many, can be water and sodium metal, or gunpowder and flint, acetone peroxide, etc)
- **Picric Acid Doped Bombs:** all the regular bombs can be made more powerful by adding some picric acid to their recipe.
Crafted: the bomb and picric acid.